MA1521 Cheat Sheet, AY25/26 S2

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1 Mathematical Formulae

1.1 Algebra

Factoring Formulae

$$a^2 - b^2 = (a+b)(a-b)$$
$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$
$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

Quadratic Formulae

Number of roots of quadratic function

$$b^2 - 4ac = 0 \text{ one-root}, > 0 \text{ two-root}, < 0 \text{ no-root}$$

Inflection point of quadratic function:

$$x = -\frac{b}{2a}$$

Completing the square formula:

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right)$$

1.2 Trigonometry

Definitions

$$\sec x = \frac{1}{\cos x} \qquad \tan x = \frac{\sin x}{\cos x}$$
$$\csc x = \frac{1}{\sin x} \qquad \cot x = \frac{\cos x}{\sin x}$$

Basic Identities

$$\sin(-x) = -\sin x \qquad \sin^2 x + \cos^2 x = 1$$
$$\cos(-x) = \cos x \qquad 1 + \tan^2 x = \sec^2 x$$
$$\tan(-x) = -\tan x \qquad 1 + \cot^2 x = \csc^2 x$$

Compound Angle Formulae

$$\sin(x+y) = \sin x \cos y + \cos x \sin y \qquad \tan(x+y) = \frac{\tan x. + \tan y}{1 - \tan x \tan y}$$
$$\sin(x-y) = \sin x \cos y - \cos x \sin y$$
$$\cos(x+y) = \cos x \cos y - \sin x \sin y \qquad \tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$
$$\cos(x-y) = \cos x \cos y + \sin x \sin y$$

Double Angle Formulae

$$\sin 2x = 2 \sin x \cos x \qquad \cos 2x = \cos^2 x - \sin^2 x$$
$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x} \qquad = 2\cos^2 x - 1$$
$$= 1 - 2\sin^2 x$$

Formulae for Reducing Power

$$\sin^2 x = \frac{1 - \cos 2x}{2} \qquad \sin^3 x = \frac{1}{4}(3 \sin x + \sin 3x)$$
$$\cos^2 x = \frac{1 + \cos 2x}{2} \qquad \cos^3 x = \frac{1}{4}(3 \cos x + \cos 3x)$$

Hyperbolic Functions

$$\sinh x = \frac{1}{2}(e^x - e^{-x}) \qquad \cosh^2 x - \sinh^2 x = 1$$
$$\cosh x = \frac{1}{2}(e^x + e^{-x}) \qquad 1 - \tanh^2 x = \operatorname{sech}^2 x$$
$$\coth^2 x - 1 = \operatorname{csch}^2 x$$
$$\tanh x = \frac{\sinh x}{\cosh x}$$

1.3 Differentiation

Standard Derivatives

$$\frac{d}{dx}(x^n) = nx^{n-1} \qquad \frac{d}{dx}(\sec x) = \sec x \tan x$$
$$\frac{d}{dx}(\ln x) = \frac{1}{x} \qquad \frac{d}{dx}(\csc x) = -\csc x \cot x$$
$$\frac{d}{dx}(a^x) = a^x \ln a \qquad \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}(e^x) = e^x \qquad \frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}(\sin x) = \cos x \qquad \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$
$$\frac{d}{dx}(\cos x) = -\sin x \qquad \frac{d}{dx}(\sinh x) = \cosh x$$
$$\frac{d}{dx}(\tan x) = \sec^2 x \qquad \frac{d}{dx}(\cosh x) = \sinh x$$
$$\frac{d}{dx}(\cot x) = -\csc^2 x \qquad \frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

Rules of Differentiation

Let $u = u(x), v = v(x), y = y(u)$

- **Sum Rule**

$$\frac{d}{dx}(u+v) = u' + v'$$

- **Quotient Rule**

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - v'u}{v^2}$$

- **Product Rule**

$$\frac{d}{dx}(uv) = u'v + v'u$$

- **Chain Rule**

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

1.4 Integrations

Standard Integrals

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \qquad \int \csc^2 x dx = -\cot x + C$$
$$\int \frac{1}{x} dx = \ln|x| + C \qquad \int \sec x \tan x dx = \sec x + C$$
$$\int a^x dx = \frac{a^x}{\ln a} + c \qquad \int \csc x \cot x dx = -\csc x + C$$
$$\int e^x dx = e^x + C \qquad \int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$
$$\int \sin x dx - \cos x + C \qquad \int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln\left|\frac{x-a}{x+a}\right| + C$$
$$\int \cos x dx = \sin x + C \qquad \int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C$$
$$\int \tan x dx = -\ln|\cos x| + C \qquad \int \frac{1}{\sqrt{a^2 + x^2}} dx = \ln\left|x + \sqrt{x^2 + a^2}\right| + C$$
$$\int \cot x dx = \ln|\sin x| + C \qquad = \sinh^{-1}\left(\frac{x}{a}\right)$$
$$\int \sec x dx = \ln|\sec x + \tan x| + C \qquad \int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln\left|x + \sqrt{x^2 - a^2}\right| + C$$
$$\int \csc x dx = -\ln|\csc x + \cot x| + C \qquad \cosh^{-1}\left(\frac{x}{a}\right) + C$$
$$\int \sec^2 x dx = \tan x + C \qquad \int \frac{1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a} \sec^{-1}\left|\frac{x}{a}\right| + C$$

2 Limits

(Two-sided) Limit L is defined as the value of f as $x \to x_0$(but $x \neq x_0$)

$$\lim_{x \to x_0} f(x) = L \text{ where } L \in \{\mathbb{R}, \infty, -\infty\}$$
$$\Leftrightarrow \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L$$

2.1 Continuity and Discontinuity

$f(x)$ is **continuous** at $x = x_0$ when

$$\lim_{x \to x_0} f(x) = f(x_0)$$

Else, the function is says to be discontinuous for one of the following reasons:

- **Removable Discontinuity**

$$\lim_{x \to x_0} f(x) = \lim_{x \to x_0^+} f(x) = \lim_{x \to x_0^-} f(x) \neq f(x_0)$$

E.g. $\frac{\sin x}{x}$ is undefined at $x = 0$, but its lim exists at $x = x_0$

- **Jump Discontinuity**

$$\lim_{x \to x_0^+} f(x) \neq \lim_{x \to x_0^-} f(x)$$

- **Infinite Discontinuity**

$$\lim_{x \to x_0^+} f(x) = \pm\infty \vee \lim_{x \to x_0^-} f(x) = \pm\infty$$

- **Other (ugly) discontinuities**

Limit does not exists when the function changes every tiny bit of changes in x (e.g. wiggle swigging line)

2.1.1 Theorem: Differentiable Implies Continuous

If f is differentiable at x_0, then f is continuous at x_0.

2.2 Techniques for Evaluating Limits

Limits of Trigonometry

$$\lim_{x \to 0} \frac{\sin x}{x} = 1, \lim_{x \to \infty} \frac{\sin x}{x} = 0$$
$$\lim_{x \to 0} \frac{1 - \cos\theta}{\theta} = 0$$

Limits of Exponential Functions

$$\lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e$$
$$\lim_{n \to \infty} \left(1 + \frac{2}{n}\right)^{5n} = \lim_{m \to \infty} \left(1 + \frac{2}{2m}\right)^{5 \cdot 2m} = e^{10}$$

L'Hôpital's Rule

- Simplify limits in indeterminate form (i.e. $\frac{0}{0} \vee \frac{\infty}{\infty}$)

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}$$

3 Differentiation

First Principles of Differentiation

Derivative of f evaluated at $x = x_0$:

$$\lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

3.1 Methods of Differentiation

Implicit Differentiation

- By differentiating w.r.t. x on both LHS and RHS, we may find $f'(x)$ without isolating the function (which is hard sometimes).

$$y^n = x^m \Leftrightarrow ny^{n-1} \frac{dy}{dx} = mx^{m-1} \Leftrightarrow \frac{dy}{dx} = \frac{mx^{m-1}}{ny^{n-1}}$$

Exponential Function

$$\frac{d}{dx}(x^x) = \frac{d}{dx}\left(e^{\ln(x^x)}\right) = \frac{d}{dx}\left(e^{x \ln(x)}\right) = x^x(1 + \ln x)$$

Differentials

$$dy = f'(x)dx; \frac{dy}{dx} = \frac{f(x)}{g(x)}; g(x)dy = f(x)dx; \int g(x)dy = \int f(x)dx$$

3.2 Applications of Differentiation

Optimization of Functions / Min-Max Problems

Given function f, x is **critical point**(s) of a function when $\boldsymbol{f'(x) = 0}$

- Critical point is **minimum when $\boldsymbol{f''(x) > 0}$** ; **maximum when $\boldsymbol{f''(x) < 0}$**

Second derivative test

Given $f(x, y)$, compute the determinant of **Hessian Matrix** to determine **properties of critical points**.

$$\text{Hessian Matrix} = \det\left(\begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}\right)$$
$$= \begin{cases} < 0 \mid \text{saddle} \\ > 0 \mid f_{xx} > 0 \to \text{local min}, f_{xx} < 0 \to \text{local max} \\ = 0 \mid \text{undetermined} \end{cases}$$

Newton's Methods

Iteratively, we may **approximate the root of the function**, through finding the **x-intercept of tangent line**.

$$y - f(x_n) = f'(x_n)(x_{n+1} - x_n) \quad \text{(Eqn of Tangent Line)}$$
$$(0) - f(x_n) = f'(x_n)(x_{n+1} - x_n) \quad \text{(At x-intercept, y=0)}$$
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Mean-Value Theorem

If f is differentiable on (a, b), and continuous on $[a, b]$, then

$$\frac{f(b) - f(a)}{b - a} = f'(c), (\text{for some } c, a < c < b\ )$$

Gradient of tangent line (c) = Gradient of secant line (a,b)

We may use MVT to approximate values of a function

$$f(x) = f(a) + f'(c)(x - a), a < c < x$$

Or Proof an inequalities where f'>0 means an increasing function.

Tangent Plane

Given f_x and f_y are slopes function of two lines tangent to the graph.

Equation of the tangent plane to the graph is

$$(z - z_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Lagrange Multipliers

Given functions f, g, where $g(x, y, z) = C$ and $f(x, y, z)$ attains maximum / minimum, there exists $\lambda \in \mathbb{R}$

$$\nabla f = \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix} = \lambda \begin{pmatrix} g_x \\ g_y \\ g_z \end{pmatrix} = \lambda \nabla g$$

4 Integrations

Anti-Derivatives

$$F(x) = \int f(x)dx$$

Where F is anti-derivative of f, or in another words $F'(x) = f(x)$

Definite Integrals and Riemann Sum

$$\int_a^b f(x)dx = \lim_{n \to \infty} \sum_{k=1}^{n} f(x_k)\Delta x = \lim_{n \to \infty} \sum_{k=1}^{n} f\left(a + k\left(\frac{b-a}{n}\right)\right) \frac{b-a}{n}$$

Fundamental Theorem of Calculus

Part 1: (The Antiderivatives) If f is continuous on $[a, b]$, $F(x) = \int_a^x f(t)dt$ is continuous on $[a, b]$ and differentiable on (a, b), and its derivative is $f(x)$

$$F'(x) = \frac{d}{dx} \int_a^x f(t)dt = f(x)$$

Part 2: (The Evaluation Theorem) If f is continuous over $[a,b]$ and F is antiderivative of f on $[a,b]$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

Example FTC1:

$$F(x) = \int_0^{x^2} \sqrt{t}\sin t dt, \text{Let } \; G(u) := \int_0^u \sqrt{t}\sin t dt, F(x) = G(u = x^2)$$

$$F'(x) = G'(u) * 2x = 2x * \sqrt{u}\sin u = 2x^2\sin(x^2)$$

Improper Integrals

$$\int_a^\infty f(x)dx = \lim_{M\to\infty}\int_a^M f(x)dx$$

4.1 Methods of Integration

Method of Substitution

Substituting the complicated expression with u to simplify integration problem.

$$u = x^4 + 2; du = 4x^3dx, dx = \frac{1}{4x^3}du$$

$$\int x^3(x^4+2)^5 dx = \frac{1}{4}\int u^5 du = \frac{u^6}{4(6)} = \frac{u^6}{24}$$

Integration By Parts

$$\int uv'dx = uv - \int u'vdx$$

Simplify integration involving **product of two function**, where:

- u simplifies after differentiating
- v' is known after integrating

Partial Fraction & Long Division

Rational function, defined ratio of polynomials $P(x)$, $Q(x)$.

When degree P < degree Q, rational functions turns into **partial fractions**

- With Q factorized to $(x-a)...(x-b)$ with $a,...,b$ are distinct:

$$\frac{P(x)}{Q(x)} = \frac{P(x)}{(x-a)...(x-b)} = \frac{c_a}{x-a} + ... + \frac{c_b}{x-b}$$

1. To find c_a, multiply both side with $(x-a)$ and take $\lim x \to a$

$$\lim_{x\to a}(x-a)\frac{P(x)}{(x-a)...(x-b)} = \lim_{x\to a}(x-a)\frac{P(x)}{...(x-b)} = \frac{P(x)}{...(x-a)} = c_a$$

- With Q factorized to $(x-a)^n(x-b)$ with a,b distinct

$$\frac{P(x)}{Q(x)} = \frac{P(x)}{(x-a)^n(x-b)} = \frac{c_{a_1}}{x-a} + \frac{c_{a_2}}{(x-a)^n}... + \frac{c_b}{x-b}$$

1. To find c_{a_2}, multiply both side with $(x-a)^n$ and take $\lim x \to a_2$

$$\lim_{x\to a_2}\left((x-a)^n\frac{P(x)}{(x-a)(x-a)^n...(x-b)} = \frac{P(x)}{(x-a)...(x-b)}\right) = c_{a_2}$$

2. Find c_b using same cover-up method as above
3. Find c_{a_1} by substituting $x = 0$ (or other dummy x to evaluate)

When degree P ≥ degree Q, we may use **long division** to split the polynomials into form of 1 + quotient polynomial.

Example:

$$\frac{x^2+x+1}{x^2+8x} = \frac{1*(x^2+8x)+(-7x+1)}{x^2+8x} = 1 + \frac{-7x+1}{x^2+8x}$$

	1
x^2+8x	x^2+x+1
	−7x+1

4.2 Applications of Integration

Area under curve

Area bounded between f(x) and g(x) from a to b.

$$A = \int_{x=a}^{x=b} f(x) - g(x)dx$$

Volume of Solids of Revolution

$$V = \int_{x=a}^{x=b} A(x) \text{ (area) } dx \text{ (width)}$$

$$\text{By x-axis } V_x = \int_{x=a}^{x=b}\pi y^2 dx = \int_{y=c}^{y=d} 2\pi y f(y)dy$$

$$\text{By y-axis: } V_y = \int_{y=c}^{y=d}\pi x^2 dy = \int_{x=a}^{x=b} 2\pi x f(x)dx$$

Average Value

Average value of y_{avg} over interval (a,b) is evaluated as:

$$y_{\text{avg}} = \frac{1}{b-a}\int_a^b ydx$$

Arc length

Arc length of curve $y = f(x)$ from point $A = (a, f(a))$ to point $B = (b, f(b))$ is evaluated as:

$$L = \sum_i \sqrt{(\Delta x_i)^2 + (\Delta y_{\text{id}})^2} = \int_{x=a}^{x=b}\sqrt{1+(f'(x))^2}dx = \int_{t=d}^{t=c}\sqrt{x'(t)^2+y'(t)^2}dt$$

Surface area revolved around axis

Area of surface generated by revolving graph of $y = f(x)$ about x-axis is

$$S_x = \int_a^b 2\pi f(x)\sqrt{1+(f'(x))^2}dx = \int_{t=a}^{t=b} 2\pi y(t)\sqrt{x'(t)^2+y'(t)^2}dt$$

and surface generated about y-axis is

$$S = \int_c^d 2\pi x\sqrt{1+(g'(y))^2}dy$$

4.3 Techniques for Double-Integrals

Exchanging order of integration

Double integral may be easier by integrating in different order. Sketch the region, then reverse the limits by finding the new endpoints from the diagram.

When both endpoint are constant, direct swapping is possible.

$$\int_{x=0}^{x=1}\int_{y=x}^{y=\sqrt{x}}\frac{e^y}{y}dydx = \int_{y=0}^{y=1}\int_{x=y^2}^{x=y}\frac{e^y}{y}dxdy$$

Change of Variables

To simplify a double integral for integrand or bounds of integration, after taking account Scale Factor, J.

$$\text{Jacobian, } J = \frac{\partial(u,v)}{\partial(x,y)} = \det|\begin{pmatrix} u_x & u_y \\ v_x & v_y\end{pmatrix}|$$

$$\therefore dxdy = \frac{1}{|J|}dudv.$$

Area, Volume, Mass

$$\text{Area of Region R} = \int\int_R 1dA = \int\int_R \sqrt{1+z_x^2+z_y^2}dxdy$$

$$\text{Mass per unit area, M} = \int\int_R \delta dA(\delta \text{ is density})$$

$$\text{Average value of f over R} = \frac{1}{\text{Area}}\int\int fdA$$

$$\text{Center of Mass} = (\overline{x}, \overline{y}) = \left(\frac{1}{\text{mass}}\int\int_R x\delta dA, ...\right)$$

Polar Coordinates

Polar coordinates $P(r,\theta)$ are useful to represents circles, where θ is angle from x-axis, and r is its radius. Useful for circles.

$$x = r\cos\theta, y = r\sin\theta$$

$$r = \sqrt{x^2+y^2}, \tan\theta = \frac{y}{x}$$

$$dA_{(x,y)} = rA_{(r,\theta)}$$

Cylindrical coordinates

Cylindrical coordinates $P(r,\theta,z)$ is extension of polar-coordinate, with z as third coordinates. Useful for cylinder.

$$dV_{(x,y,z)} = rV_{(r,\theta,z)}$$

Spherical Coordinates

Spherical coordinates $P(\rho,\theta,\Phi)$ where ρ is length from origin, θ is angle in x-y plane, from x-axis. Φ is angle from +ve z-axis. Useful for sphere.

$$z = \rho\cos(\Phi); r = \rho\sin(\Phi)$$

$$dV_{(x,y,z)} = \rho^2\sin\Phi V_{(\rho,\theta,\Phi)}$$

5 Infinite Series and Convergence Tests

5.1 Geometric and Harmonic Series

Geometric Series

$$S = \lim_{n\to\infty}\sum_{i=1}^n ar^{i-1} = a + ar + ar^2 + ar^3 + ...$$

$$r(a + ar + ar^2 + ar^3 + ...) = ar + ar^2 + ar^3 + ... = rS$$

$$(a + ar + ar^2 + ar^3 + ...) - (ar + ar^2 + ar^3 + ...) = S - rS = a$$

$$\therefore S \text{ converges to } \frac{a}{1-r} \text{ iff} - 1 < a < 1 \text{ else diverges}$$

P-Series / Harmonic Series

$$S = \sum_{n=1}^\infty \frac{1}{n^p} \sim = \int_0^\infty \frac{1}{x^p}dx$$

Converges:

$$\text{When p>1} := \lim_{a\to\infty}\int_0^a \frac{1}{x^p}dx = \lim_{a\to\infty}\frac{x^{-p+1}}{-p+1}|_{x=0}^{x=a} = \frac{1}{p-1} \text{ as -p+1<1}$$

Diverges:

$$\text{When p=1} := \lim_{a\to\infty}\int_0^a \frac{1}{x}dx = \lim_{a\to\infty}\ln a = \infty$$

$$\text{When p<1} := \lim_{a\to\infty}\int_0^a \frac{1}{x^p}dx = \lim_{a\to\infty}\frac{x^{-p+1}}{-p+1}|_{x=0}^{x=a} = \infty \text{ as -p+1>1}$$

5.2 Convergent Tests

Divergent Test

$$\lim_{n\to\infty} a_n \neq 0 \vee \infty \Rightarrow \text{Series Diverge, else unknown (further test required)}$$

Ratio Test

Determine if a function converges absolutely by comparing the ratio of the current term and next term.

$$p = \lim_{n\to+\infty}|\frac{a_{n+1}}{a_n}|, p = \begin{cases}\text{converge absolutely} \mid p < 1 \\ \text{diverge} \mid p > 1 \\ \text{unknown} \mid p = 1 \;\; \text{(further test required)}\end{cases}$$

Comparison Test

Given two series a_n, b_n where $\forall n, a_n \leq b_n$

- If b_n converges absolutely, a_n converges
- If a_n diverges, b_n diverges

$$S_a := \sum_{k=1}^\infty \frac{1}{n+1}, S_b := \sum_{k=1}^\infty \frac{n}{n+1}$$

Since $a_n < b_n$ and S_a diverges, thus S_b diverges by Comparison Test

Integral Test

$$f \geq 0 \text{ decreasing } \wedge f(n) = a^n$$

$$\int_1^\infty f(x)dx \text{ converge} \Rightarrow \sum_{n=1}^\infty a_n \text{ converge}$$

$$\int_1^\infty f(x)dx \text{ diverge} \Rightarrow \sum_{n=1}^\infty a_n \text{ diverge}$$

Alternating Series Test

- By Leibniz Alternating series test, an alternating series ($(-1)^k$) converges when:
  - $|a_n|$ decreases monotonically, (i.e. $|a_{n+1}| < |a_n|$) and
  - $\lim_{n\to\infty} a_n = 0$

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + ... \to \text{ converges}$$

Absolute and Conditional Convergence

- Series converges absolutely if $\sum|a_k|$ converge
- Series converges conditionally if $\sum a_k$ converge but $\sum|a_k|$ diverge

5.3 Taylor's series and Power Series

Power Series

- **Power Series** consist of powers of variable x in the form

$$\sum_{k=0}^\infty c_k x^k = c_0 + c_1x + c_2x^2 + ...$$

- **Radius of Convergence, R** for power series refers to value of x such that the power series converge when $x \in (-R, R)$.
  - If the series diverge, Radius of Convergence is 0.

Taylor's Series

- **Taylor's Series** approximates a function by finding the relevant coefficient to the power series polynomials.

$$f(x) = \sum_{n=0}^\infty a_n x^n = a_0x^n + a_1x_1 + ... + a_nx_n + ...$$

$$\text{where } a_n = \left(\frac{1}{n!}\right)f^{(n)}(0)$$

- Taylor Series at x=a

$$f(x) = f(b) + \frac{f'(b)}{1!}(x-b) + \frac{f''(b)}{2!}(x-b)^2 + ...$$

5.4 Common Taylor Series

$$\frac{1}{1-x} = \sum_{k=0}^\infty x^k = 1 + x + x^2 + x^3 + ...; -1 < x < 1$$

$$\frac{1}{1+x^2} = \sum_{k=0}^\infty (-1)^k x^{2k} = 1 - x^2 + x^4 - x^6 + ...; -1 < x < 1$$

$$e^x = \sum_{k=0}^\infty \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + ...; -\infty < x < +\infty$$

$$\sin x = \sum_{k=0}^\infty (-1)^k\frac{x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + ...; -\infty < x < +\infty$$

$$\cos x = \sum_{k=0}^\infty (-1)^k\frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + ...; -\infty < x < +\infty$$

$$\ln(1+x) = \sum_{k=1}^\infty (-1)^{k+1}\frac{x^k}{k} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ...; -1 < x \leq 1$$

$$\tan^-1(x) = \sum_{k=0}^\infty (-1)^k\frac{x^{2k+1}}{2k+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + ...; -1 \leq x \leq 1$$

$$\sinh(x) = \sum_{k=0}^\infty \frac{x^{2k+1}}{(2k+1)!} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + ...; -\infty < x < +\infty$$

$$\cosh(x) = \sum_{k=0}^\infty \frac{x^{2k}}{(2k)!} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + ...; -\infty < x < +\infty$$

$$(1+x)^m = 1 + \sum_{k=1}^\infty \frac{m(m-1)...(m-k+1)}{k!}x^k; -1 < x < 1$$